

Your Index Is Fine, Your Query Encoder Is Not: Query Distillation over Frozen Indexes

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Abstract

Document embeddings are reused across searches, allowing the cost of a large document encoder to be amortized, while every uncached query incurs an encoding cost. This asymmetry motivates replacing the query encoder with a smaller model that searches the existing document embeddings. Such a replacement can reduce query compute while avoiding the re-embedding and index migration required by a change of document encoder. We examine how the existing index shapes the retrieval quality and search cost of this replacement.

We fit two types of smaller query encoder (students) across 26 embedding models: a learned token-vector table and a small frozen transformer with a learned projection. Their retrieval quality is associated with both the original model’s quality and its embedding dimensionality. At a given dimensionality, stronger original models tend to yield stronger students; after accounting for original-model quality, higher dimensionality predicts weaker students. Together, these properties predict student quality on held-out models better than original-model quality alone.

The encoding savings also depend on how the student’s queries interact with approximate search. Token-table students require more graph-search effort to recover their exact nearest neighbors, with a corpus-dependent effect on retrieval relevance. Constella illustrates the combined trade-off: its compact transformer and token-table query encoders search the unchanged document index of the Stella embedding model, retaining 90.5% and 81.4% of Stella’s exact-search nDCG@10 on BEIR-15. Separate timing measurements show that additional search work reduces their encoding savings. The existing index therefore shapes both the quality a smaller query encoder can achieve and how much of its encoding advantage translates into cheaper retrieval.

1. Introduction

In a production retrieval system, reducing query compute by replacing the embedding model can turn an optimization into an index migration. Changing the document representation requires re-embedding the corpus and rebuilding the vector index. Downtime can be avoided, but keeping retrieval available during the transition can require old and new vector indexes to coexist and ingestion to keep both representations current until traffic switches [Index migration]. Backfill and index construction then add compute, storage, and I/O demands alongside live queries and updates. The migration also needs relevance validation, a coordinated query-encoder switch, and a rollback path. Its duration and cost depend on corpus token volume, encoding throughput, index-building capacity, and the rate of incoming updates. Replacing only the query encoder with a compatible student preserves the stored document vectors and avoids that migration; its effects on retrieval quality and search effort still need to be measured.

Query-side distillation trains a student encoder to reproduce the frozen encoder’s query vectors and search the frozen document vectors directly [QED, EmbedDistill, LEAF]. Static variants fit a token table, so a query costs a table lookup and a sum [pyNIFE, LightRetriever]. We study the retrieval quality and search effort of these replacements, and whether properties of the existing embedding space predict their performance.

We examine two properties of the space: embedding dimensionality (the number of components in each vector) and the original encoder’s retrieval quality. A frozen index inherits both from its document encoder. Fitting each student by a linear solve takes minutes, allowing the same recipe to be compared across 26 public encoder spaces. Two questions organize the paper.

- **RQ1, retention.** How much retrieval quality can a query student fitted by a linear solve recover from a frozen index, and do embedding dimensionality and the space’s own quality predict it? We test the project’s initial heuristic that a better space yields a better student, using exact retrieval.
- **RQ2, search effort.** Over the index’s unchanged approximate-search graph, how much more search effort does the student’s query path need than the index’s own query path, and is that effort predictable?

Student quality follows two opposing associations: stronger spaces favor the student, while higher dimensionality predicts weaker students. A model using both properties predicts held-out and prospectively tested encoders (Section 4). The token-table students also need more graph effort to recover their exact neighbors in every measured space on two of three corpora. The distance between queries and documents predicts the size of that penalty, which also appears in LightRetriever’s jointly trained lookup path (Section 5).

Constella is the worked instance: a frozen Stella index [Stella] with three compatible query paths, Stella’s own encoder, a 34.5-million-parameter transformer student (Nano), and a token-table student (Zero), released with the registered evaluations that gated the release. Section 6 reports it.

Throughout, “registered” marks the project’s pre-registered evaluations, whose decision rules were fixed before any result was observed, and “exploratory” marks analyses added afterwards. The two index-wide screens that answer RQ1 and RQ2 are exploratory; their analyses were written down before scoring, and the prospective test of RQ1 commits its predictions before any student is fitted.

2. Related work

Query-side distillation against a frozen document encoder is established: QED retains 92.5% of a dense retriever’s BEIR score with a two-layer student [QED], EmbedDistill distills an asymmetric student against the teacher’s frozen document encoder [EmbedDistill], and LEAF reports 97.7% retention at 4.7-fold compression [LEAF]. pyNIFE fits a static token table to a frozen teacher [pyNIFE], while LightRetriever trains a lookup query side jointly with a document tower [LightRetriever]. Static sentence embeddings, Model2Vec, and NanoVDR explore related compact representations [StaticEmb, Model2Vec, NanoVDR]. RQ1 extends this work with a common-recipe comparison across encoder spaces; RQ2 tests LightRetriever’s released lookup path over its own index alongside the fitted students.

Stronger teachers can produce weaker students in distillation generally [Cho and Hariharan 2019] and in dense retrieval [PROD]; those works discuss capacity gaps between teacher and student. Here, embedding dimensionality is a predictive property tested across encoder spaces, with held-out and prospective predictions.

Graph-based approximate search over HNSW is sensitive to where queries sit relative to the indexed vectors [HNSW]. Out-of-distribution queries can need more effort, and graph constructions use query samples to mitigate it [OOD-DiskANN, RoarGraph]. RQ2 examines query substitution over an unchanged graph. Query-aware decoding of binary sketches is the precedent for the precision control in Appendix C [QA-Cos].

3. Experimental setup

3.1 Encoder spaces

The roster is 26 public English embedding encoders that share the 30,522-entry BERT WordPiece vocabulary, a requirement of the table student. Ten were registered before any result; 16 were added as an exploratory expansion under the same recipe and selection order. Each encoder defines one frozen space: its document vectors are frozen and its own query encoder is the reference path. Table A lists them by dimensionality.

Table A. The 26 encoder spaces by dimensionality (full roster in Appendix B). “Own nDCG@10” is the encoder’s own query path on the six-set suite (Section 3.3); student columns are the outcomes of Section 4.

Dimensionality	Spaces (registered)	Own nDCG@10, range	Table student, range	Head student, range
384	9 (0)	0.328 to 0.504	0.290 to 0.358	0.196 to 0.508
768	10 (4)	0.285 to 0.533	0.186 to 0.353	0.059 to 0.325
1024	7 (6)	0.474 to 0.597	0.246 to 0.397	0.156 to 0.268

The roster spans nine related model families and three embedding sizes; several checkpoints are revisions of one another. We report checkpoint-resampling intervals alongside family-level sensitivity analyses. The arctic-embed-l mean-pooling control is excluded from roster statistics; its published CLS pooling is used for the roster entry.

3.2 Closed-form query students

The cross-model comparison uses two student representations fitted by ridge regression to the original encoder’s query vectors. This gives each space the same inexpensive fitting procedure while comparing a static token representation with contextual transformer features. The original encoders and document vectors stay frozen throughout.

Table student. A 30,522-row table assigns one learned vector to each WordPiece token. A query is represented by the mean of its token vectors, including repeated tokens, followed by L2 normalization. Ridge regression fits the rows jointly, with a penalty toward vectors obtained by running individual tokens through the original encoder. Only the table is fitted; query encoding then consists of tokenization, lookups, and pooling.

Head student. The pretrained bge-small transformer [BGE] provides contextual features. We mean-pool the token states from layers 12, 8, and 4 using the attention mask and concatenate them into a 1,152-dimensional vector. A learned linear projection with bias maps this vector to the document embedding dimensionality, then L2 normalization produces the query vector. The backbone stays frozen, so fitting learns only the projection. Appendix B specifies both regression objectives and their solvers.

These fitted students support the comparison across encoder spaces. The released Zero and Nano models use additional optimization and different training data, described below; their measured performance is reported in Section 6.

3.3 Fit data, selection, and scoring

Both students are fitted on the same 337,981 screened fit queries, encoded by each index’s own query path with its published prompt. The ridge penalty is chosen per index on two development forums (CQADupStack physics and programmers) and frozen before any public dataset is scored. Retrieval is scored by exact search, nDCG@10 with the registered self-hit drop, on six public BEIR datasets [BEIR]: NFCorpus, SCIDOCS, SciFact, TREC-COVID, FiQA, and ArguAna. The six-set macro is the outcome; the four sets without disclosed exposure for the Stella instance (the first four) are a sensitivity outcome. Each index is scored once per set.

3.4 Approximate-search protocol

RQ2 uses Qdrant 1.19.1 [Qdrant 1.19.1] with uncompressed vectors, cosine distance, and HNSW graphs ($m=16$, $ef_construct=100$). We build two graphs per space and workload using different seeded insertion orders. The workloads are FiQA (57,638 documents), SCIDOCs (25,657), and TREC-COVID (171,332). Each graph is searched with the original encoder’s queries and both students’ queries, sweeping ef , which controls the breadth of graph search, from 16 to 512.

Each path is compared with its own exact top-10. The pre-specified primary measure is the relative-nDCG-loss multiplier: the smallest student ef that reaches the original path’s relative loss at $ef=64$, divided by 64 and averaged over builds. A student that fails to reach that reference by $ef=512$ in a build is reported as censored, with that build’s multiplier at least eight. The secondary measures are the analogous multiplier for tie-aware recovery@10, the fraction of exact neighbors recovered, and the recovery gap at $ef=64$. Appendix C gives the parity checks, tie rule, and tas-b exclusion.

3.5 Constella: separately trained query encoders for one index

Constella supplies two query encoders for the 1024-dimensional document embeddings of `stella_en_400M_v5` [Stella]. Stella supplies fixed training targets and encodes the documents used for retrieval. Zero and Nano learn different query representations compatible with that space.

Zero: a trained token table. Zero learns one vector and one scalar weight per WordPiece token. Its first training phase combines agreement with Stella’s query vectors and matching Stella’s score distribution over candidate documents. A second, contrastive phase trains the query vectors to rank relevant documents above negatives, using frozen Stella document vectors. At export, scalar weights are folded into the table rows and the rows are quantized to int8. The served encoder pools token vectors with square-root count weights: a token occurring c times contributes $\sqrt{c}$ times its stored vector. The pooled vector is L2-normalized. This representation gives each token the same vector in every query, trading contextual computation for lookup and pooling.

Nano: a jointly optimized transformer and projection. Nano starts from the 12-layer bge-small transformer and concatenates the 384-dimensional token states from layers 12, 8, and 4. A linear projection with bias maps the resulting 1,152 features to Stella’s 1,024 dimensions; attention-masked mean pooling and L2 normalization produce the query vector. The projection is initialized by ridge regression, then the backbone and projection are trained together to minimize squared L2 error against normalized Stella embeddings. Training alternates three query batches with one document batch. Queries use Stella’s query targets and documents use its document targets; retrieval uses Stella’s stored document vectors. The resulting model has 34.5 million parameters.

Both models were selected through development evaluations before their benchmark results were scored. Their training sources, objectives, optimization settings, and selection procedures are specified in Appendix B. Zero’s learned weights, contrastive training, and served pooling differ from the fitted table; Nano updates the backbone that the fitted head keeps frozen. These differences define the scope of the cross-model findings and the separate Constella evaluation.

4. RQ1: what the index decides about retention

4.1 Hypothesis and the pooled result

H1. The stronger the index’s own retrieval, the stronger the closed-form student fitted to it; in particular, the strongest index yields the strongest student.

H1 was the project’s initial selection rule. The strongest registered space, gte-large-en-v1.5, yields the weakest registered table student and a below-median head student. Stella, second in the same dimensionality band, yields the strongest table student. Across all 26 spaces, the rank correlation between the original encoder’s score and its student’s score is +0.09 for the table (95% checkpoint interval -0.39 to $+0.52$) and +0.09 for the head (-0.36 to $+0.51$). These wide intervals allow a moderate positive association. Section 4.4 evaluates the selection rule by the quality of the student it picks.

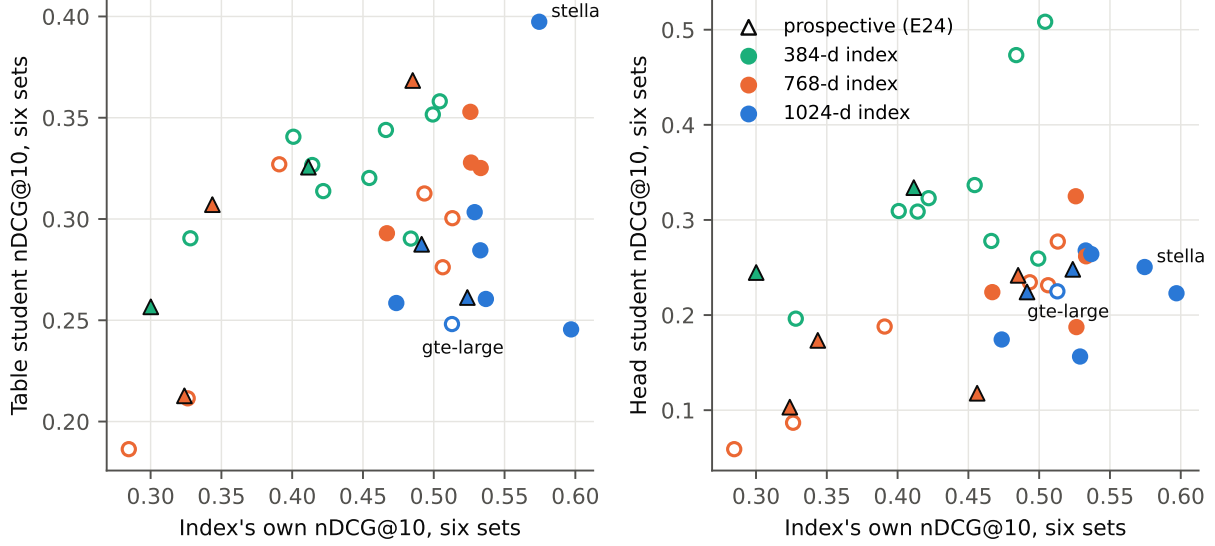


Figure 1: Student quality against the space’s own quality for the table (left) and head (right) students, colored by dimensionality. Filled points are registered spaces; triangles are the prospective spaces of E24 (eight for the head, seven for the table).

4.2 Dimensionality as the competing explanation

H1’. At a given dimensionality, stronger spaces yield stronger students; after accounting for space quality, higher-dimensional embeddings predict weaker students. Support requires a dimensionality coefficient whose interval excludes zero and better held-out-family prediction than space quality alone.

Table B models the student’s absolute six-set score using the space’s own score and log2 embedding dimensionality. Retention is the ratio of student to original-encoder scores. Standardized coefficients express the change in student quality, in standard deviations, per standard deviation of a predictor. Intervals use 10,000 checkpoint resamples. Leave-one-family-out prediction fits on eight model families and predicts the ninth.

Table B. Nested models of student quality over 26 indexes (exploratory, analysis pre-specified).

Student	Predictors	Space quality beta	Dimensionality beta	In-sample R ²	Leave-one-family- out Spearman
Head	space quality	+0.34 [−0.16, +0.65]		0.12	−0.17
Head	space quality, dimensional- ity	+0.69 [+0.40, +0.97]	−0.83 [−1.14, −0.60]	0.70	+0.69
Table	space quality	+0.37 [−0.30, +0.72]		0.14	−0.15
Table	space quality, dimensional- ity	+0.63 [+0.06, +0.89]	−0.64 [−0.95, −0.32]	0.48	+0.40

Space quality and dimensionality correlate at +0.61 across the roster. Their opposing associations with student quality help explain the near-zero pooled rank correlation. Given dimensionality, the partial rank correlation between student and index quality is +0.58 for the head and +0.45 for the table. Table B shows a larger standardized coefficient for dimensionality under the head representation and coefficients of similar

magnitude under the table representation. Family-adjusted fits and coefficient intervals are reported in Appendix B.

4.3 Held-out and prospective prediction

We test whether the quality-and-dimensionality model ranks further encoders better than their own retrieval scores do. Fitted on the ten registered spaces, the model ranks the 16 later spaces at Spearman 0.85 [0.53, 0.99] for the head and 0.70 [0.20, 0.96] for the table; the original scores alone rank them at 0.37 and 0.16. The hypothesis was formed after all 26 spaces had been seen, making this a held-out fit.

For the prospective test, we chose eight previously unscored BERT-vocabulary encoders spanning the three embedding sizes, fitted the model on the 26, and committed its predictions by hash before fitting any student (Appendix B). One table fit stopped at the convergence gate, leaving seven table students and eight head students. The committed predictions rank them at Spearman +0.86 [+0.33, +1.00] for the head and +0.71 [−0.12, +1.00] for the table, against +0.26 and +0.39 for the original scores alone. Mean absolute error of the predicted six-set score is 0.039 (head) and 0.032 (table). The head’s checkpoint interval supports a positive association; the table’s wider interval leaves its prospective result uncertain.

4.4 Corollary: choosing an index before it exists

RQ1 concerns a space that already exists. For a team that can still choose, the same data answer a second question: which rule picks the encoder whose student will be best? Table C scores three rules by the six-set nDCG@10 their pick loses against the best student in the roster.

Table C. Six-set nDCG@10 lost against the best student in the roster, by selection rule.

Rule	Head: pick, regret	Table: pick, regret
Strongest space (H1)	gte-large-en-v1.5, 0.285	gte-large-en-v1.5, 0.152
Two-variable model, leave-one-out pick	arctic-embed-s, 0.249	gte-small, 0.107
Student’s own dev-forum screen	gte-small, 0.035	stella-400M-v5, 0.000
Prospective spaces (eight head, seven table): strongest space	0.086	0.107
Prospective spaces: committed model pick	0.000	0.043
Prospective spaces: dev screen	0.000	0.043

The student’s development screen makes the best or near-best pick under both representations. It scores the fitted student on two forums and predicts six-set rank at Spearman 0.84 (head) and 0.88 (table). Adding it to the quality-and-dimensionality model gives it the largest standardized coefficient (+0.71 [+0.36, +0.99] for the head) and raises leave-one-family-out prediction to 0.77 and 0.80. The quality-and-dimensionality model captures the ranking structure, while testing a fitted student improves selection. On the prospective spaces, the model’s committed pick matches the development screen’s; selecting by the original encoder’s score incurs the largest regret on both rosters.

The head’s backbone, bge-small-en-v1.5, is also a roster encoder. Removing it changes every statistic above by less than 0.06.

5. RQ2: what the index decides about search effort

5.1 Hypothesis

H2. Over the index’s unchanged HNSW graph, the student’s query path needs more search effort than the index’s own query path to recover the same fraction of its exact neighbors, because the student’s queries sit differently against the indexed vectors.

We first measure the effort penalty per index, then test whether query-placement features predict its size (Section 5.3).

5.2 Effort across 25 indexes

H2 predicts a multiplier above one in most spaces on every workload. Table D tests it.

Table D. Search effort of the table student relative to the index’s own path at **ef=64**, two builds averaged, 25 indexes (exploratory, analysis pre-specified). The primary quantity is the relative-nDCG-loss multiplier; “censored” means at least one build never reached the reference by **ef=512**.

Workload	Documents	Loss multiplier: indexes above 1 / at or below	Censored	Loss multiplier median over reached	Recovery gap at ef=64, mean (indexes below 0)	Recovery multiplier median
FiQA	57,638	25 / 0	4	4.0	−2.9 pp (25 of 25)	4.0
SCIDOCS	25,657	17 / 8	8	1.25	−0.8 pp (17 of 25)	2.0
TREC-COVID	171,332	12 / 13	8	0.25	−3.9 pp (25 of 25)	4.0

The primary measure holds on FiQA (25 of 25, median four over the 21 reached, four censored in at least one build) and on SCIDOCS (17 of 25). It is mixed on TREC-COVID (12 of 25): over 50 queries, the index’s own path loses only 0.2% of nDCG at **ef=64**, so the relative-loss threshold there is sensitive to differences smaller than its noise. By the secondary measure it holds on every workload (Table D), and the head student shows the same recovery deficit in 24 of 25 spaces on FiQA and 23 of 25 on TREC-COVID.

H2c predicts a smaller recovery gap on a smaller corpus. On FiQA subsampled to SCIDOCS’s 25,657 documents, with the same queries and two rebuilt graphs per space (exploratory, pre-specified; Appendix C), the gap at **ef=64** moves from −2.9 points to −1.2, a paired change of +1.8 points (95% interval over spaces +1.3 to +2.3). It remains 0.4 points larger than SCIDOCS’s (−0.9 to −0.1). The subsample reproduces much of SCIDOCS’s attenuation, with a remaining difference between the corpora.

5.3 Is the effort predictable?

H2b predicts that query-placement features computed before graph construction rank the spaces by their recovery gap. We declared nine features and two covariates before computing them (exploratory, pre-specified; Appendix C). Table F focuses on the query-to-document distance ratio: the distance from queries to their nearest documents, relative to the typical distance between neighboring documents. Correlations use 75 space-by-workload points, with intervals resampling each space together with its three workload rows. Held-out prediction fits a ridge model on all families except one and predicts that family.

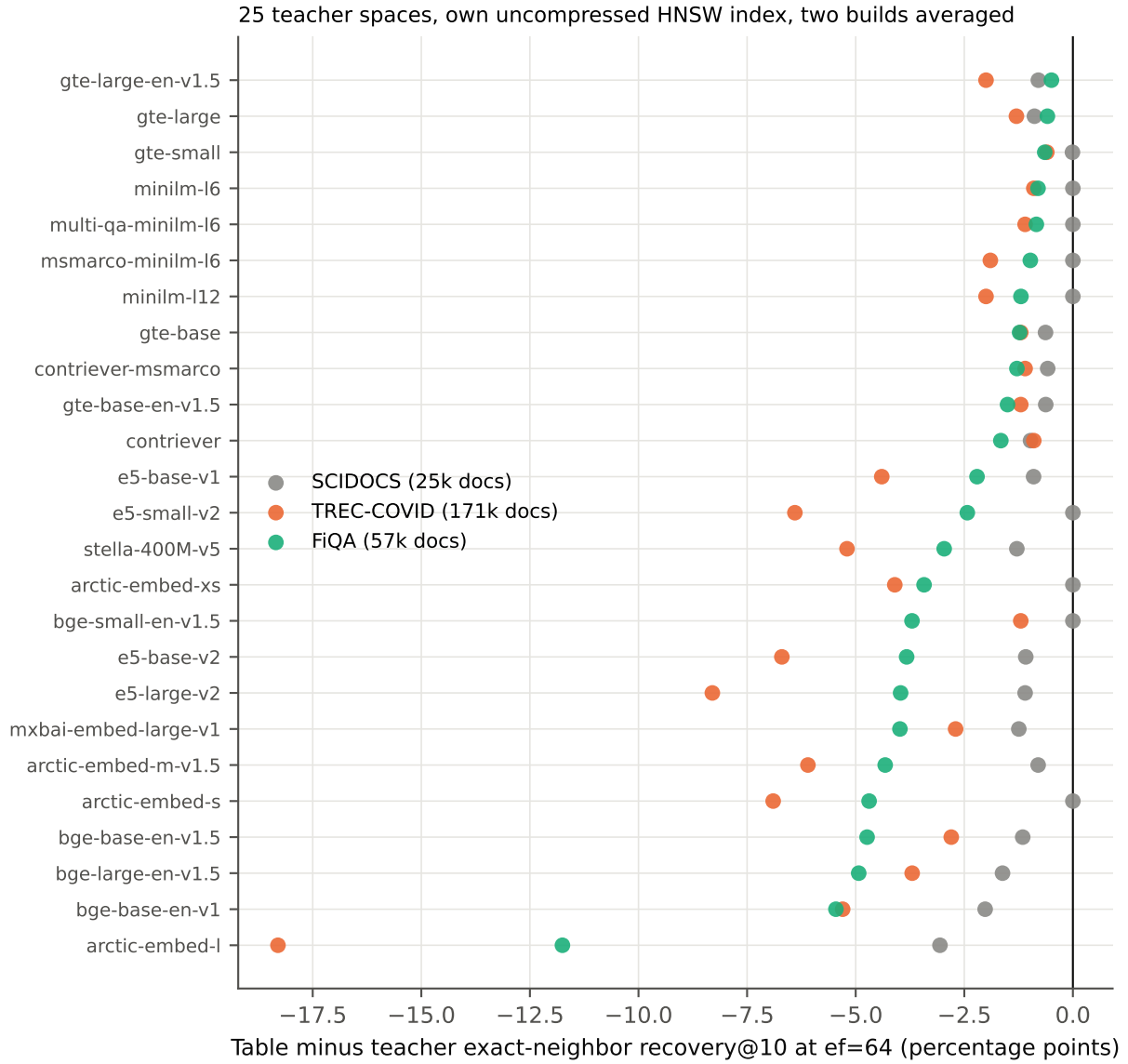


Figure 2: Recovery gap at **ef=64** by index, three workloads, two builds averaged.

Table F. Predictors of the recovery gap over 75 points (25 spaces by three workloads). Negative Spearman means a larger feature value goes with a larger penalty.

Feature	Spearman, 75 points	95% interval	FiQA / SCIDOCS / TREC-COVID	Leave-one-family- out Spearman, MAE (pp)
Query-to- document distance ratio, encoder’s queries	−0.64	[−0.75, −0.48]	−0.80 / −0.32 / −0.83	+0.64, 1.26
Query-to- document distance ratio, student’s queries	−0.63	[−0.74, −0.46]	−0.77 / −0.45 / −0.69	+0.56, 1.79
All nine features and two covariates				+0.73, 1.40
Training-fold mean (baseline)				2.22

Spaces with queries farther from documents tend to pay a larger recovery penalty. The ratio computed from the original encoder’s queries predicts held-out families with lower absolute error than the all-feature model (Table F), so the existing query path already supplies a useful predictor. The association is clearest on FiQA and TREC-COVID; on SCIDOCS its interval includes zero. Appendix C reports the remaining features, including changes between student and original-encoder queries.

Prediction across workloads is less precise. A model fitted on two workloads ranks the third at +0.50, but its mean error is 2.55 points against 2.33 for the training-fold mean. The ratio therefore supports relative ranking, with the magnitude of the penalty requiring calibration on the target workload.

5.4 The penalty in a jointly trained model

H2d. If the deficit comes from where lookup queries sit against a graph built for the encoder’s own queries, it should also appear when the lookup side is trained jointly with the document tower rather than fitted afterwards. We tested the released LightRetriever model (Qwen2.5-1.5B with its published adapter) over its own FiQA and SCIDOCS document vectors, comparing its lookup query path with the same model’s full query encoding under the Section 3.4 protocol, with the model’s web-search instruction as the primary prompt and the task prompt as secondary. Predictions and quantities were fixed before encoding (exploratory, pre-specified).

Table E. LightRetriever’s lookup path against its full query path over its own index, two builds averaged.

Workload	Prompt	Exact nDCG@10, lookup / full	Loss multiplier (builds)	Recovery multiplier	Recovery gap at ef=64, query 95% interval
FiQA	web search	0.4065 / 0.4752	3.0 (2, 4)	4.0	−3.6 pp [−4.3, −2.9]
FiQA	task	0.4124 / 0.4766	2.0 (2, 2)	4.0	−2.3 pp [−2.9, −1.8]
SCIDOCS	web search	0.1655 / 0.1925	2.25 (0.5, 4)	4.0	−1.7 pp [−2.1, −1.4]
SCIDOCS	task	0.1818 / 0.2017	censored (1, censored)	2.0	−0.7 pp [−1.0, −0.5]

The lookup path recovers fewer of its own exact neighbors at **ef=64** on both corpora under both prompts, with query intervals clear of zero. Matching the full path’s recovery takes four times the **ef** in every FiQA build and under the primary SCIDOCS prompt. The relevance-loss multiplier agrees on FiQA (two to four); on SCIDOCS, both paths lose under 0.1% of nDCG at **ef=64**, making that reference sensitive to very small differences. GPU encoding took 0.10 ms for the lookup path and 0.90 ms for the full path. These encoding timings and graph-effort multipliers measure separate components; Section 6 combines encoding and search timing for Constella.

The **ef** multipliers measure graph-search effort. Relevance loss and exact-neighbor recovery give complementary views of the substitution, with different conclusions on TREC-COVID and on LightRetriever’s SCIDOCS index.

6. The Constella instance

Table 1 gives the three query paths over Stella’s frozen index, by exact search over BEIR-15.

Table 1. Three query paths over one frozen Stella index. Encoding medians: 100 real 5-to-12-word queries, batch one, four CPU threads, Apple M5 Pro, registered protocol.

Query path	Exact nDCG@10	Share of Stella	Encode p50	Assets
Stella query encoder	0.5614	reference	31.6 ms	1669.6 MiB
Nano (34.5M transformer)	0.5081	90.5%	2.25 ms	132.3 MiB
Zero (token table)	0.4572	81.4%	0.044 ms	90.1 MiB

All three query paths were verified against the same populated collections, allowing each request to choose a compatible encoder. Appendix A gives the per-dataset results and selected-reference comparisons.

Build cost and query cost. Table-student fitting took four to seven minutes of A100 time per encoder, including query encoding except for Stella’s cached targets. Nano’s final optimization took 57.3 A100 hours, about \$95 at the recorded rental rate, excluding data preparation, recipe search, and failed runs. Appendix B gives the component costs. Both constructions preserve the existing document index.

Table 2 extrapolates registered timing medians to encoding-plus-search time per million queries on the measurement laptop. Encoding uses batch one with four threads; search uses the uncompressed million-passage collection at each path’s setting within 1% of its own exact score. These are sums of separately measured encoding and search times.

Table 2. Extrapolated encoding-plus-search time per million queries on the measurement laptop, from the medians in Table 1 and Appendix C.

Query path	Encoding	Search, within 1% of own exact	Total	Share of Stella
Stella query encoder	8.78 h	0.43 h (ef =128)	9.2 h	reference
Nano	0.63 h	0.61 h (ef =256)	1.2 h	13%
Zero	0.01 h	0.99 h (ef =512)	1.0 h	11%

Search time at a fixed **ef** differs by less than 5% across paths. The extra work comes from the larger **ef** needed to reach each path’s relative-quality target: 128 for Stella, 256 for Nano, and 512 for Zero. Zero’s extra search largely consumes its encoding advantage over Nano, bringing their extrapolated encoding-plus-search times close together (Table 2). Their exact BEIR-15 quality is reported separately in Table 1. On one binary-quantized collection, the 63-fold encoding gap between Zero and Nano becomes 2.2-fold after search, at different quality levels (Figure 3). Appendices C and D report query-precision controls, lexical fusion, and per-query routing over the same index.

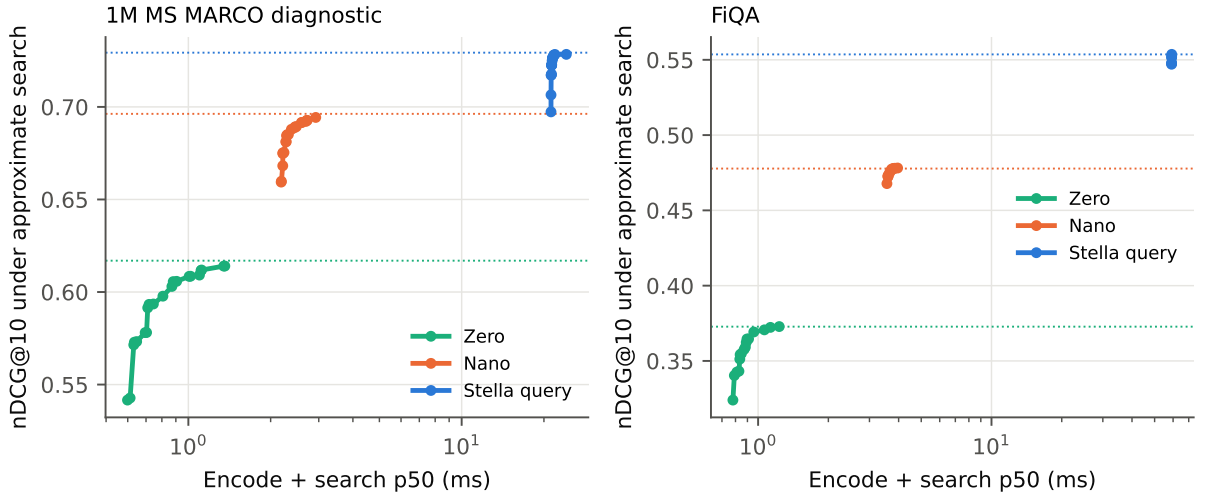


Figure 3: Quality under approximate search against encoding plus search time, registered sweep. Dotted lines are each path’s exact score.

7. Limitations

RQ1 covers two closed-form recipes over 26 related spaces at three embedding sizes. Dimensionality covaries with family and quality, making its association predictive rather than causal. Checkpoint-resampling intervals describe variation within this roster; family-level prediction tests transfer across its families. Transfer to fully trained students such as Zero and Nano remains an open question.

RQ2 covers one engine, one graph configuration, and three workloads with different sizes and domains, including 50 queries on TREC-COVID. Its predictors establish associations with search effort; cross-space timing ran on a shared machine, so latency conclusions rely on the separate Constella sweep. The LightRetriever replication covers one released 1.5-billion-parameter model, our reproduction of its dense path, and two corpora.

Constella’s students use different training pools, including different FEVER exposure, and Stella discloses exposure on ArguAna, FiQA, and FEVER. Query intervals condition on the trained models. Three bounded

specialization attempts on an internal workload produced no improved encoder; per-workload students remain a direction for future research.

Artifact availability

The companion repository holds inference examples, build configurations, methods with their dated amendments, immutable result receipts, per-dataset tables, and the full research history including negative results. The released artifacts are Constella Zero, Constella Nano, and the Stella document encoder on the Hugging Face Hub. Two inputs are identified by hash but not yet archived in the repository: the cleaned 337,981-query fit list and the sampled one-million-passage identifier list. Prepared Zero and Nano training corpora are described by source and file manifests rather than bundled as raw text.

Appendix A. Registered evaluations of the instance

Serving configuration. Stella is pinned to revision `ffeb2b7e`, with its `s2p_query` prompt, fp32 encoding, normalized output, fp16 stored document vectors, and cosine similarity; maximum sequence length 512. Zero’s training pool includes FEVER-train and HotpotQA; Nano excludes FEVER. MS MARCO is licensed for non-commercial use, so it was validation-only in our distillation: it supplied no training targets, negatives, or generation seeds, although Nano’s pretrained base has prior exposure to it.

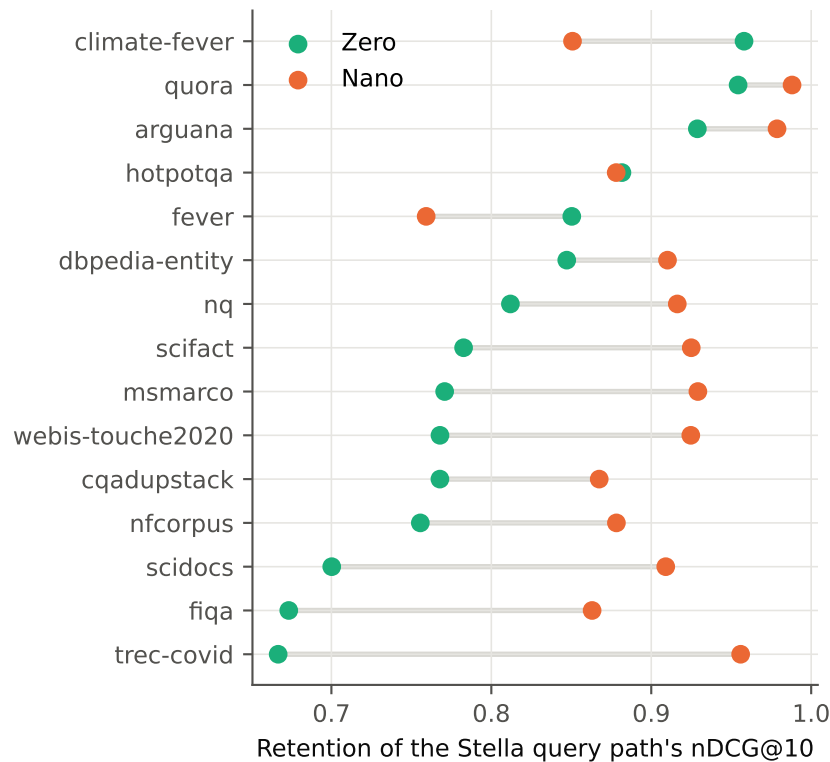


Figure A1: Retention per dataset. Datasets with disclosed Stella training exposure are ArguAna, FiQA, and FEVER; the four originally held-out sets are FEVER, DBpedia-entity, and the CQADupStack android and english forums.

Variation across datasets. Zero’s sum of token vectors is insensitive to word order and context. Its retention is 0.67 on TREC-COVID and FiQA, against 0.95 on Quora. Nano’s retention ranges from 0.85 to 0.99 except FEVER, which was in Zero’s training pool and excluded from Nano’s. These dataset differences describe the released models and their distinct training pools.

Partitions. The six-set suite is NFCorpus, SCIDOCs, SciFact, TREC-COVID, FiQA, and ArguAna. “Clean-four” means the first four, which have no disclosed Stella overlap; it is not a causal contamination estimate. Four datasets were held out and evaluated once: FEVER, DBpedia-entity, and the CQADupStack android and english forums. Two other forums, physics and programmers, select table penalties and routing settings and were reused during model construction.

Reference systems. bge-small-en-v1.5 and LEAF were selected for the project’s original evaluations. Each uses its own document representation. On BEIR-15 they score 0.5171 and 0.5402 against Nano’s 0.5081, at 2.57 and 1.39 ms median encoding; BM25 scores 0.4006. Nano’s broad-average score is below both dense references; the registered partition-specific contrasts follow.

Nano’s registered contrasts used a fixed sequence with one-sided 2.5% lower bounds. “Established” means the bound cleared zero; “unresolved” means it did not; “descriptive” rows were registered as reports, not tests. Intervals resample queries with the models fixed.

Contrast	Partition	nDCG@10 difference	Outcome
Nano – bge-small	clean-four	+0.0176; lower bound +0.0037	established
Nano – bge-small	all six	+0.0274; lower bound +0.0173	established
Nano – LEAF	all six	+0.0162; lower bound +0.0065	established
Nano – LEAF	clean-four	−0.0011; lower bound −0.0145	unresolved
Nano – bge-small	held-out NDO-3, dataset weighted	+0.0032 [−0.0069, +0.0134]	descriptive
Nano – bge-small	held-out NDO-3, query pooled	−0.0121 [−0.0219, −0.0024]	descriptive
Nano – LEAF	held-out NDO-3	−0.0389 [−0.0488, −0.0290]	descriptive

NDO-3 is the registered held-out aggregate: DBpedia-entity weighted one half, the android and english forums one quarter each, FEVER excluded as the double-exposure sensitivity row.

Zero’s registered contrasts used a Holm correction across the family. Zero missed its dense bar and did not confirmatorily beat BM25.

Contrast, all six	nDCG@10 difference	Raw 95% interval	Outcome
Zero – LightRetriever dense	−0.0243	[−0.0405, −0.0086]	below the 0.4583 bar
Zero – BM25	+0.0165	[+0.0017, +0.0311]	unresolved; p=.0149 fails Holm .0083
Zero + BM25 (convex) – OpenSearch	+0.0043	[−0.0063, +0.0151]	unresolved

The registered clean-four sensitivity is descriptive: −0.0443 [−0.0675, −0.0212], −0.0311 [−0.0517, −0.0109], and −0.0107 [−0.0262, +0.0043]. Unresolved superiority does not establish equivalence. The confirmatory fusion operator was convex score fusion with development-fitted weight 0.8 at depth 1000; the later DBSF-at-100 recommendation is a deployability choice, not a new confirmation. On the development components DBSF minus convex is +0.0035 at 10 candidates, −0.0020 at 50, −0.0063 at 100, and −0.0146 at 1000.

The held-out four, copied from their one-shot published aggregate:

System	FEVER	DBpedia	CQADup android	CQADup english
Nano	0.6231	0.4190	0.4774	0.4298
Zero	0.6978	0.3900	0.4431	0.3690
Stella	0.8207	0.4603	0.5275	0.5183

System	FEVER	DBpedia	CQADup android	CQADup english
bge-small	0.8662	0.4003	0.4761	0.4558
LEAF	0.8671	0.4520	0.5260	0.4707
BM25	0.5036	0.3045	0.3952	0.3481
Zero + BM25	0.7559	0.4030	0.4629	0.4016
Nano + BM25	0.6879	0.4296	0.4750	0.4323

Precision. All released paths run in fp32 through the published loader; a candidate fp16 export passed CPU parity and failed on CUDA (minimum cosine 0.662), which is why qualification runs through the actual loader and device.

Appendix B. Construction details

Fitted table: objective and preprocessing. Let X contain each query’s token counts divided by its token count, Y the original encoder’s query vectors, and W_0 its normalized outputs for individual tokens encoded between the tokenizer’s special tokens. We fit the table W by

$$\min_W \|XW - Y\|_F^2 + \lambda \|W - W_0\|_F^2.$$

The initialization and regularization anchor therefore use the encoder’s complete pooling and output projection. Tokenization includes special tokens, uses no student query prefix, and truncates at 512 tokens. The student mean-pools table rows and L2-normalizes the result. Block conjugate gradient solves the normal equations; every output column must pass the relative-residual tolerance (1e-6 in fp32, 1e-8 in fp64). The penalty grid is 1e-4 to 1e-1, extended by one decade once when the best development value is at an edge. The chosen value maximizes exact-search nDCG@10 on the two development forums and is frozen before six-set scoring. The fit list was cleaned after an earlier version had 1.31% held-out-query overlap. One expanded-roster checkpoint failed the convergence gate and is excluded. Registered checkpoint exposure is documented; the exploratory expansion was not individually audited.

Fitted head: objective and features. Frozen bge-small at revision 5c38ec7c supplies attention-masked mean states from layers 12, 8, and 4. Concatenating them with a constant bias feature gives the design matrix F with 1,153 columns. A dense solve fits A to the same targets Y :

$$\min_A \|FA - Y\|_F^2 + \lambda \frac{\text{tr}(F^\top F)}{1153} \|A\|_F^2.$$

The bias is regularized along with the other coefficients, and the prediction FA is L2-normalized for retrieval. The penalty grid, development criterion, and selection order match the table’s. The backbone stays fixed throughout. bge-small-en-v1.5 is also an index in the roster and shares the backbone, and gte-small is nearly its linear image (retention 1.008 and 0.978); correlations without the backbone are +0.09 against the index and +0.46 against the table.

Released Zero: data and initialization. Zero’s training uses 338,076 usable query-document pairs plus 220,632 query-only rows from Amazon ESCI, FEVER-train, HotpotQA-train, SQuAD-train, Mr. TyDi English, NQ Open, and TriviaQA. These are distinct from the cleaned 337,981-query list used for the fitted students. The distillation phase also uses 924,704 short spans drawn from the document pool as pseudo-queries. Each of the 30,522 table rows starts from Stella’s normalized output for that token in a short context; a learned scalar token weight starts from inverse document frequency. Stella’s weights, query targets, and document vectors stay frozen, while table rows and token weights are optimized.

Zero’s two training phases. The first phase runs for 16,000 steps with batches of 512 and ordinary count pooling. Half of each batch is drawn from the combined query-only and pseudo-query pool and receives cosine alignment loss. The paired-query half combines cosine alignment with KL divergence from Stella’s

softmax score distribution over a positive document and 31 distractors sampled from a document-vector bank. The extra-text cosine mean is added separately to the paired-query loss. The cosine and KL weights are 1 and the score temperature is 0.02. Adam uses learning rates 3e-3 for rows and 1e-2 for token weights. A row penalty of 1e-3 pulls toward initialization, divided by one plus that row’s update count.

The second phase starts from that checkpoint and runs 2,500 contrastive steps on the paired queries. InfoNCE scores the positive against 32,768 sampled bank negatives plus dataset-provided negatives, at temperature 0.02. Known positives are masked from the negatives; Stella also masks a candidate whose score exceeds the positive’s score minus 0.02. Adam uses peak learning rates 1e-3 for rows and 1e-2 for weights, with 200 warmup steps followed by linear decay to a tenth of those rates. The row penalty is anchored to the starting checkpoint and both phases use seed 0. The bank contains 2,000,000 frozen document vectors.

Zero’s selection and served representation. Development retrieval selected the final two-phase recipe and, after training with mean pooling, selected square-root count pooling for the unchanged table. Export folds the learned scalar weights into the rows and applies per-row absolute-maximum int8 quantization with fp32 scales. At inference, a token occurring c times contributes $\sqrt{c}$ times its dequantized row before L2 normalization. Rows never updated in either phase remain at their initialization. The development suite comprises NQ, HotpotQA, two CQADupStack forums, and two held-out training-source slices; it was reused during recipe selection. Its training-adjacent components and adaptive reuse motivate reporting the benchmark evaluations separately (Appendix A).

Released Nano: data and objective. Nano’s deduplicated pool contains 3,486,034 query texts: 403,443 real queries from ESCI, HotpotQA, Mr. TyDi, NQ Open, SQuAD, and TriviaQA; 999,744 PAQ questions; 1,248,385 harvested texts; and 834,462 generated queries. Harvested claims, titles, and keywords are drawn from Wikipedia, arXiv metadata, and the licensed document pool. Queries are sampled with replacement to give equal presentation shares to 12 forms: factoid, how-to, claim, argument, finance, title, keyword, health, product, comparison, yes/no, and conversational. The 6,070,049 eligible training documents are replayed in reshuffled epochs. FEVER is excluded from both pools. Queries have no student prefix and document examples receive `passage::`; targets use Stella’s corresponding query or document encoding. The training objective is mean squared L2 distance between normalized student and target vectors, with three query batches followed by one document batch. The document examples train Nano, while Stella remains the document encoder used at retrieval.

Nano’s generated queries and filtering. The generation recipe uses Qwen3-8B-AWQ at revision 4da05a8e, with thinking disabled, temperature 0.8, top-p 0.95, and request seeds derived from passage identifiers. The seven generated forms are how-to, argument, finance, health, comparison, yes/no, and conversational. Finance and health queries use Wikipedia body passages; the other five forms use passages from the HotpotQA corpus. Form-specific prompts request five queries per passage; the reply must parse as a JSON list, with one retry on failure. Token budgets are 60 per requested query, or 400 for argument and conversational forms. Assembly applies form-length checks, exact deduplication, document-based hold-out exclusions, query and document overlap screens, and rejection of copied spans of five words. Forms with a near-duplicate rate above 25% retain only representative queries; a form with fewer than 50,000 survivors is dropped. A seeded uniform sample caps each generated form at 143,000 queries before corpus-wide deduplication. The companion repository provides the prompts, source revisions, filtering rules, and manifests identifying the resulting texts.

Nano’s initialization, optimization, and selection. The bge-small backbone is pinned to revision 5c38ec7c. Its layers 12, 8, and 4 supply 1,152 features per token to a biased linear projection into 1,024 dimensions, followed by masked mean pooling and normalization. A ridge warm start fits the projection on 60,000 training queries (seed 21), using a 50,000/10,000 split to select the penalty by normalized squared L2 error from 1e-6 to 1; the selected penalty is 0.001 and the projection is refitted on all 60,000 queries. A linear projection commutes with mean pooling, so this pooled-feature initialization gives the same query computation as applying the projection before pooling in the served model.

The backbone and projection then train jointly with AdamW: batch 32, seed 0, betas (0.9, 0.999), epsilon 1e-8, weight decay 0.01 on matrices and zero on one-dimensional parameters, and gradient clipping at norm 1. CUDA forward passes use bf16 and the loss uses fp32. Three cycles decay linearly from 1e-4 to 1e-5,

with a 64,000-example warmup in the first cycle. The released model is the third cycle-end checkpoint, after 199,999,721 example presentations. Recipe screening and scheduled monitoring use a development relevance macro that weights four families equally: six BRIGHT slices, MedicalQARetrieval, LEDGER, and the LegalBench CorporateLobbying and ConsumerContractsQA datasets. Scores are averaged within each family first. Registered screening rules selected the recipe, retaining defaults when comparisons were unresolved. The companion repository records the data manifests, generation prompts, source revisions, selection rules, and released artifact hashes.

The roster. The 26 encoder spaces of Section 3.1, by dimensionality and own six-set score.

Encoder (frozen space)	Dimensionality	Pooling	Own nDCG@10	Table student	Head student	Roster
bge-small-en-v1.5	384	cls	0.5042	0.3581	0.5084	exploratory
arctic-embed-s	384	cls	0.4993	0.3516	0.2594	exploratory
gte-small	384	mean	0.4838	0.2903	0.4733	exploratory
arctic-embed-xs	384	cls	0.4662	0.3440	0.2779	exploratory
e5-small-v2	384	mean	0.4544	0.3203	0.3367	exploratory
minilm-l12	384	mean	0.4219	0.3138	0.3229	exploratory
minilm-l6	384	mean	0.4142	0.3267	0.3088	exploratory
multi-qa-minilm-l6	384	mean	0.4008	0.3406	0.3094	exploratory
msmarco-minilm-l6	384	mean	0.3281	0.2905	0.1962	exploratory
gte-base-en-v1.5	768	cls	0.5331	0.3252	0.2618	registered
arctic-embed-m-v1.5	768	cls	0.5263	0.3279	0.1874	registered
bge-base-en-v1.5	768	cls	0.5259	0.3529	0.3249	registered
bge-base-en-v1	768	cls	0.5131	0.3004	0.2773	exploratory
gte-base	768	mean	0.5063	0.2762	0.2314	exploratory
e5-base-v1	768	mean	0.4934	0.3126	0.2345	exploratory
e5-base-v2	768	mean	0.4669	0.2930	0.2241	registered
contriever-msmarco	768	mean	0.3909	0.3270	0.1879	exploratory
tas-b	768	cls	0.3262	0.2115	0.0868	exploratory
contriever	768	mean	0.2846	0.1864	0.0590	exploratory
gte-large-en-v1.5	1024	cls	0.5970	0.2455	0.2229	registered
stella-400M-v5	1024	mean	0.5745	0.3974	0.2505	registered
mxbai-embed-large-v1	1024	cls	0.5368	0.2605	0.2641	registered
bge-large-en-v1.5	1024	cls	0.5329	0.2845	0.2679	registered

Encoder (frozen space)	Dimensionality	Pooling	Own nDCG@10	Table student	Head student	Roster
arctic-embed-l	1024	cls	0.5289	0.3034	0.1564	registered
gte-large	1024	mean	0.5129	0.2481	0.2249	exploratory
e5-large-v2	1024	mean	0.4735	0.2585	0.1743	registered

Family-adjusted quality model. Adding eight family indicators to the quality-and-dimensionality model raises in-sample R^2 to 0.91 (head) and 0.81 (table), preserving both coefficient signs. These models describe the observed families and are reported in sample. In Table B, the head’s standardized dimensionality and quality coefficients are -0.83 and $+0.69$; the table’s are -0.64 and $+0.63$. The table coefficient intervals overlap in magnitude.

Published retention context. LightRetriever reports about 95% retention for its jointly trained lookup path, while Zero retains 81.4% over frozen Stella here. These figures describe different models and evaluation protocols. Their comparison leaves the quality cost attributable to freezing the index unresolved.

Prospective roster (E24). Eight encoders fixed in the method before any encoding, all passing the tokenizer check: e5-large (1024), bge-large-en v1 (1024), e5-base-unsupervised (768), nomic-embed-text-v1 (768), msmarco-bert-base-dot-v5 (768), msmarco-distilbert-base-v4 (768), all-MiniLM-L12-v1 (384), paraphrase-MiniLM-L6-v2 (384). Teacher scores were measured first, the prediction file was written and its sha256 recorded (81f2c88e...) before any student was fitted, and every student command verified that hash. The table fit for e5-base-unsupervised stopped at the convergence gate at the extended penalty $1e-5$; its head fit completed. E24 amendment 1 includes each completed fit, giving eight head points and seven table points. Per encoder, six-set score of the space, then table student actual / predicted, then head student actual / predicted:

Encoder	Dimensionality	Own nDCG@10	Table: actual / predicted	Head: actual / predicted
bge-large-en v1	1024	0.5237	0.2613 / 0.2883	0.2481 / 0.2103
e5-large	1024	0.4914	0.2875 / 0.2757	0.2243 / 0.1825
nomic-embed-text-v1	768	0.4851	0.3684 / 0.2945	0.2418 / 0.2336
e5-base-unsupervised	768	0.4562	stopped / 0.2833	0.1178 / 0.2087
all-MiniLM-L12-v1	384	0.4115	0.3256 / 0.3171	0.3340 / 0.3062
msmarco-distilbert-base-v4	768	0.3436	0.3071 / 0.2395	0.1734 / 0.1117
msmarco-bert-base-dot-v5	768	0.3239	0.2127 / 0.2318	0.1032 / 0.0947
paraphrase-MiniLM-L6-v2	384	0.3001	0.2566 / 0.2738	0.2447 / 0.2103

msmarco-bert-base-dot-v5 is the teacher QED distilled from; its closed-form students retain 66% (table) and 32% (head) of its six-set score here, against the 92.5% QED reports for its trained two-layer student on BEIR under its own protocol, which the comparison does not reproduce.

Correlation rosters. Registered ten: Spearman head-versus-index $+0.18$ (-0.56 to $+0.67$), head-versus-table $+0.24$ (-0.49 to $+0.74$), head development-versus-public $+0.71$ ($+0.12$ to $+0.96$). Leave-one-family-out values range from $+0.44$ to $+0.60$ for head-versus-table and -0.12 to $+0.27$ for head-versus-index over the

26. Within dimensionality bands, head-versus-index Spearman is +0.47 $[-0.47, +1.00]$ at 384 (n=9), +0.67 $[-0.03, +1.00]$ at 768 (n=10), and +0.36 $[-0.76, +0.87]$ at 1024 (n=7).

Vector fidelity and retrieval quality. Tables fitted to two e5 checkpoints match their index’s query vectors at mean cosine 0.90 and 0.89 against 0.78 for Stella’s table, yet retrieve worse. Three earlier training screens reduced their losses while leaving ranking unimproved; three vector-error diagnostics also failed to rank table quality. These observations motivate selection by retrieval against the target document vectors.

Costs. Table fitting four to seven minutes per registered index on an A100, including query encoding except cached Stella targets; head features 43 seconds for the fit list, dev grid 14 seconds per index, six-set scoring about one minute per index. Nano’s final optimization ran 57.3 hours on one A100 for 199,999,721 example presentations, about \$95 at the recorded rental rate, excluding target preparation, data generation, recipe search, failed runs, and evaluation. Zero’s retraining was estimated at 20 minutes with prepared targets, plus 8 to 12 hours to encode targets for a new index.

Appendix C. Search and precision controls

Instance sweep. Qdrant 1.19.1, HNSW m=16, ef_construct=100, indexing threshold 1 KB so no segment is served by a plain scan, ef in 16 to 512, oversampling 1, 2, and 4 under compression, exact parity against NumPy on 200 queries per path. Timing is warmed and sequential with encoding and search measured in separate phases. At ef=16 the uncompressed million-passage losses are 4.4%, 7.0%, and 16.3% for Stella, Nano, and Zero; under binary quantization without oversampling 6.1%, 9.2%, and 19.8%.

Cross-index sweep. The same engine and graph parameters, uncompressed, two builds per index and workload differing in a seeded insertion order, 16 to 512 ef, limit 11 with the self-hit drop, tie-aware exact parity on 200 queries per path. Four amendments were recorded during the run, before any index’s rows were analysed: SCIDOCS added as a third workload because TREC-COVID has 50 queries; the tie-aware measure; a 0.995 sanity gate on parity with all values reported; and the exclusion rule that removed tas-b (SCIDOCS head parity 0.994). Stella’s own-path rows agree with the instance sweep at every ef. Loss-based censoring: four indexes on FiQA, eight on SCIDOCS, eight on TREC-COVID; recovery-based censoring: one per workload.

Truncated queries (registered). On synthetic prefixes of SciFact, NFCorpus, and FiQA queries, the three dense paths retain descriptively similar shares of their own full-query nDCG@10 at each prefix length; this is not an equivalence test and no search-as-you-type session was measured.

Gap predictors (E22). Features per space and workload, declared before computation: the mean cosine distance from a query to its nearest document divided by the mean nearest-neighbor distance among a 5,000-document sample (for the student’s and the encoder’s queries, and their difference); the Gini coefficient of document occurrence counts across the student’s exact top-10 lists over the full document support, minus the encoder’s; the mean overlap of the student’s and encoder’s exact top-10; the effective rank (participation ratio) of a seeded 20,000-row sample of the encoder’s fit-query vectors and the ratio of the student’s to the encoder’s workload-query effective rank; the student-minus-encoder top-1 cosine and margin recorded by the search sweep; log2 dimensionality and log10 corpus size as covariates. Ridge with unit penalty on standardized features; the baseline is each training fold’s mean. The seven features not in Table F, same columns:

Feature	Spearman, 75 points	95% interval	FiQA / SCIDOCS / TREC-COVID	Leave-one-family- out Spearman, MAE (pp)
Student minus encoder ratio	−0.23	[−0.42, −0.03]	−0.37 / −0.44 / +0.17	−0.15, 2.23
Hubness of the student’s top-10, minus the encoder’s	+0.40	[+0.20, +0.57]	+0.12 / −0.20 / −0.15	+0.14, 2.11

Feature	Spearman, 75 points	95% interval	FiQA / SCIDOCS / TREC-COVID	Leave-one-family- out Spearman, MAE (pp)
Student-encoder top-10 overlap	+0.37	[+0.14, +0.60]	−0.01 / +0.43 / +0.28	+0.11, 2.29
Effective rank of the encoder’s fit-query cloud	−0.17	[−0.40, +0.11]	−0.28 / −0.34 / −0.05	−0.53, 2.38
Ratio of the student’s to the encoder’s workload-query effective rank	+0.18	[−0.06, +0.41]	+0.16 / −0.48 / +0.32	−0.30, 2.17
Top-1 cosine, student minus encoder	−0.05	[−0.30, +0.20]	0.00 / +0.23 / −0.34	−0.55, 2.28
Margin, student minus encoder	−0.28	[−0.48, −0.05]	−0.19 / −0.05 / −0.37	−0.11, 2.15

Corpus-size subsample (E23). FiQA documents drawn uniformly with seed 20260930 to 25,657, keeping every document with a positive judgment for a test query; each path’s exact top-10 and tie thresholds recomputed over the subsample; two builds; the same **ef** grid; per-space gaps paired against the full-corpus and SCIDOCS values with 10,000-draw bootstrap intervals over spaces.

Native binary pilot and replication. On one fixed binary collection per workload with 192 deterministically selected queries, binary scoring with rescoring at 1x oversampling loses 11.7, 5.1, and 2.3 percentage points of exact-neighbor recovery for Zero, Nano, and Stella on FiQA at **ef=64**; the million-passage replication loses 3.9, 3.1, and 2.2, and the extra Zero-versus-Nano interval includes zero. Native rescoring also changes cross-segment candidate merging.

Graph-free precision control. Same queries, vectors, and targets; all document sign codes scored exhaustively at global candidate budgets 10, 40, and 100; expected coverage averages uniform inclusion at tied boundaries.

Workload	Path	Sign query	Full query	Gain, points
FiQA	Zero	0.8217	0.9297	+10.80
FiQA	Nano	0.9137	0.9771	+6.33
FiQA	Stella	0.9638	0.9922	+2.84
1M diagnostic	Zero	0.9010	0.9677	+6.67
1M diagnostic	Nano	0.9484	0.9833	+3.49
1M diagnostic	Stella	0.9685	0.9932	+2.47

The paired extra Zero gains over Nano are 4.46 points on FiQA (95% query interval 2.53 to 6.43) and 3.17 on the diagnostic (1.58 to 4.75). Zero starts with more misses and does not repair a larger fraction of them. Full query values are not native scalar8 encoding, and global candidate counts are not per-segment oversampling.

Appendix D. Fusion and routing over the frozen index

Lexical fusion. Fusing Zero with BM25 raises its BEIR-15 macro score from 0.4572 to 0.4933, 87.9% of Stella; Nano plus BM25 reaches 0.5110. The sign varies by workload: Zero on HotpotQA rises from 0.6127 to 0.7015, Nano on MS MARCO falls from 0.4063 to 0.3699. On the million-passage diagnostic a dense-plus-sparse collection takes 4.71 ms per query including Zero encoding against 1.11 ms for the dense binary

example. These timings compare different collection configurations. The scores use distribution-based score fusion over 100 candidates per branch with a bm25s lexical branch [DBSF]; other lexical implementations need their own comparison because their score distributions enter the normalization. The operator ordering changes with candidate depth (Appendix A).

Internal workload. Zero fell below BM25 on one internal domain-jargon workload. Three bounded attempts to specialize a student to that workload produced no improved encoder, as summarized in Section 7.

Routing and blending (registered). On 12 datasets, an oracle choosing Zero or Nano per query with relevance labels scores 0.5473 against 0.5161 for always-Nano, and matches always-Nano by sending the most beneficial 15% of queries to Nano. Query length, subwords per word, and the pooled vector norm are weak selectors (0.002 to 0.005 over random routing at matched Nano share); Zero’s post-search score margin recovers about a quarter of the oracle gain over random routing on six datasets and costs a second search for escalated queries. A development-selected 30% Zero contribution blended into Nano’s vector lowers Nano’s six-set score by 0.0051 (95% paired interval -0.0092 to -0.0011), consistent with how hard retrieval-strategy selection and query performance prediction remain [Arabzadeh et al., QPP].

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